Math 403 Homework #3, Spring 2025

Instructor: Ezra Miller

Solutions by: ...your name...

Collaborators: ...list those with whom you worked on this assignment...
(1 point for each of up to 3 collaborators who also list you) /3

Due: 11:59pm Saturday 1 March 2025

READING ASSIGNMENTS /42

13. for Thu. 20 February
    - [Treil, §6.3.3–§6.3.4] singular value decomposition
    - [Lax, Chap.7: §Norm of a Linear Map, §Spectral Radius]
    - [Lax, Chap.8: §Norm and Eigenvalues]
    - [Treil, §6.4] Applications of SVD: spectral radius, operator norm, condition number
14. for Tue. 25 February
    - [Stewart–Sun, §II.2.1] matrix norms, consistency
    - [Stewart–Sun, §IV.1 through Thm 1.3] general perturbation theorems
15. for Thu. 27 February
    - [Stewart–Sun, §IV.1.2 through Thm 1.6] Bauer–Fike theorem
    - [Lax, Appendix 7] Gershgorin's Theorem
    - [Stewart–Sun, §IV.2 through §IV.2.1] Gerschgorin theory

16–17. for Tue. 4 March and Thu. 6 March
    - [Tapp, Chap.5] Lie algebras as tangent spaces to the identity
    - [Lax, Chap.9: Matrix-valued Functions, through §Matrix Exponential]
    - [Wikipedia, *Exponential map (Lie theory)*]

EXERCISES

1. Prove that a subset of a manifold is open if and only if its preimage under every chart is an open subset of an affine space. (Note: This can be used to specify the topology on a manifold without knowing what a topological space is, since affine spaces have the usual notion of "open": what it means for a neighborhood of a point in a manifold to be open is well defined independent of which charts are used to verify openness.) /3

   Solution

   Assume that a subset of the manifold is open but its preimage is closed. We can also assume that the preimage is not empty, since the empty set is open. Then this implies that the image of a closed subset is open under a homeomorphism, which is false, since homeomorphisms preserve topological strucutre(maps open sets to open sets). Therefore the preimage of an open sets under every chart is open, likewise same reasoning implies that the image of an open subset is open.

2. Prove that the sphere $S^2 = \left\{\mathbf{x} \in \mathbb{R}^3 \,\middle|\, ||\mathbf{x}|| = 1\right\}$ is a smooth manifold. Is it a rational algebraic variety over the field $\mathbb{R}$? /3

   Solution

   Chart 1:

$$
\begin{aligned}
U_1 &= \mathbb{R}^2 \\
\pi_1 &: U_1 \hookrightarrow S^2 \\
\pi_1^{-1} &: S^2/(0,0,1) \twoheadrightarrow U_1 \simeq R^2 \\
\pi_1^{-1} &: (x,y,z) \mapsto (\frac{x}{1-z}, \frac{y}{1-z}) \\
\pi_1 &: (x,y) \mapsto (\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{-1+x^2+y^2}{1+x^2+y^2})
\end{aligned}
$$

   Chart 2:

$$
\begin{aligned}
U_2 &= \mathbb{R}^2 \\
\pi_2 &: U_2 \hookrightarrow S^2 \\
\pi_2^{-1} &: S^2/(0,0,-1) \twoheadrightarrow U_2 \simeq R^2 \\
\pi_2^{-1} &: (x,y,z) \mapsto (\frac{x}{1+z}, \frac{y}{1+z}) \\
\pi_2 &: (x,y) \mapsto (\frac{2x}{1+x^2+y^2}, \frac{2y}{1+x^2+y^2}, \frac{1-x^2-y^2}{1+x^2+y^2})
\end{aligned}
$$

   The transition functions are $\pi_1 \circ \pi_2^{-1}$ and $\pi_2 \circ \pi_1^{-1}$, This functions are differeniable, so it is a smooth manifold. I don't know what a rational algebraic variety is, but since the transition functions are rational I suppose it is a rational algebraic variety.

3. Fix a field F. Prove that the unipotent lower-triangular matrices $U^- \subseteq GL_n(F)$ map injectively to the set $\mathcal{F}\ell_n(F)$ of complete flags in F^n expressed as the set of orbits of the group B^+ acting on the right of $GL_n(F)$; that is, $U^- \hookrightarrow \mathcal{F}\ell_n(F) = GL_n(F)/B^+$. /3

   Solution

We need to prove that any lower traingular unipotent matrix corresponds to one and only one complete Flag. Let's assume the opposite, then you get two matrices M_1 and $M_2, M_1 \neq M_2$, and the complete flags induced(Assuming the standdard induciton, where v_i as the span of the union of the left i columns) by the two matrices are the same.

Look at the first column of M_1, since this has to be a linear mulitple of the first column of M_2, and since the first entries are the same and non-zero(it is 1, because it is lower-triangular and unipotent, therefore the eigenvalues 1, are on the diagonal), then the first columns are equal.

Assuming that all previous columns are equal, what about the l^{th} column. Since all the columns are linearly independent in both M_1 and M_2(they are invertible) , then you can describe l^{th} column as a unique linear combination of the 1^{st} to l^{th} column. Since the coordinate of l^{th} column in this basis is just one and everything else being 0 in both matrices, and knowing that the subspace spanned being the same, this leads us to conclude that the columns are linear combinations of each other, and since they both share the same non-zero entry, then they are the the same column. Therefore contradicitng that two different unitary matrices can describe the same complete flag.

4. Let $S_n \subseteq GL_n(F)$ be the set of permutation matrices. Prove that $\mathcal{F}\ell_n(F)$ is a rational algebraic variety with atlas $\{\pi_w : wU^- \to \mathcal{F}\ell_n(F) \mid w \in S_n\}$ naturally indexed by S_n. You will need to use that any level set of a polynomial with coefficients in F is closed. /3

Solution

Cliam #1: Any complete flag can be described as a composition of a lower triangular unipotent matrix and permutation matrix.

Since a complete flag can be represented as a sequence of a vectors that are continiously added to a subspace, take the first vector to be added or the first subspace in the chain. You can scale this vector so that one of its nonzero entries equals 1. Subtract a multiple of the first vector from the other vectors in the chain so that all entries of the vectors corresponding to the 1 in the first vector are 0.

Scale the second vector so that a nonzero entry equals 1, and subtract a multiple of it, from the other vectors(all vectors other than the first one) so that the entry corresponding to 1 in the second vector becomes 0.

Continue this process for all vectors. You can always continue this process because if at any point you find a vector that has all entries 0, then that would imply then that vector was in the span of the other vectors, which is a contradiction because the vectors are all linearly independent.

Put all the current vectors into a matrix A, then make a permutation matrix, where the i^{th} column has a one in the j^{th} place if c_i(the i^{th} column of A), had 1 chosen at the j^{th} entry.

When you multiply A with this permutation matrix you get a lower traingular unipotent matrix, since the 1^s are going to be on the diagonal and anything after that entry will become 0 from the construction. The inverse of the permutation matrix created will be our chart.

Claim #2: The transition functions between charts are composition of permutation matrices.

To get the transition function between the charts compsose the matrix with the inverse of the other matrix, which is just another permutation matrix.

Claim #3: This manifold is a rational algebraic variety.

I don't what rational algebraic varities are, but since you the composition of the matrices is rational, I suppose it is a rational algebraic variety.

5. The *standard complex structure* on $\mathbb{R}^{2n}$ is the block-diagonal $2n \times 2n$ matrix J_{2n} whose diagonal blocks are all $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Prove that $A \in GL_{2n}(\mathbb{R})$ is complex-linear if and only if A commutes with the standard complex structure: $AJ_{2n} = J_{2n}A$. /3

Solution

For $n = 1$, assume there exists some matrix A and it is commutative with the standard complex strucutre. Let's what conditions it imposes.

$$A = \begin{bmatrix} a & c \\ d & b \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & c \\ d & b \end{bmatrix} = \begin{bmatrix} a & c \\ d & b \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} -d & -b \\ a & c \end{bmatrix} = \begin{bmatrix} c & -a \\ b & -d \end{bmatrix}$$

So this means commutatitve implies,

$$a = b$$
$$-d = c$$

Which is the same coniditions for a 2×2 to be complex linear. Note this proves both sides of the iff since, commutativity and complex linearity impose the same restrictions.

Assume true for size n, now we will prove that it is true for n+1. We only have to check for complex linearity in each 2×2 block in the matrix. So if each 2×2 commutes with the standard complex structure, then the entire matrix is complex linear.

$$\begin{bmatrix} 0 & -1 & ... \\ 1 & 0 & ... \\ ... & ... & J_{2n} \end{bmatrix} A = A \begin{bmatrix} 0 & -1 & ... \\ 1 & 0 & ... \\ ... & ... & J_{2n} \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & ...0 \\ 1 & 0 & ...0 \\ ...0 & 0 & ...J_{2n} \end{bmatrix} \begin{bmatrix} A_{11} & A_{12} & ...A_{1n} \\ A_{21} & A_n & \\ \cdots & & \\ A_{1n} & & \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & ...A_{1n} \\ A_{21} & A_n & \\ \cdots & & \\ A_{1n} & & \end{bmatrix} \begin{bmatrix} 0 & -1 & ...0 \\ 1 & 0 & ...0 \\ ...0 & 0 & ...J_{2n} \end{bmatrix}$$

$$\begin{bmatrix} A_{11}^{|} & A_{12}^{|} & ...A_{1n}^{|} \\ A_{21}^{|} & A_n^{|} & \\ \cdots & & \\ A_{1n}^{|} & & \end{bmatrix} = \begin{bmatrix} A_{11}^{-} & A_{12}^{-} & ...A_{1n}^{-} \\ A_{21}^{-} & A_n^{-} & \\ \cdots & & \\ A_{1n}^{-} & & \end{bmatrix}$$

Where $A^{|} = A^{-}$, implies that the 2×2 matrix is complex linear(from the $n = 1$ case). We now see that the matrix is commutatitve if and only if every 2×2 is complex linear which iff the matrix is complex linear.

6. Prove that if $\lambda \in \mathbb{C}$ is an eigenvalue of a unitary matrix then $|\lambda| = 1$. /3

Solution

$$U^*U = UU^* = I$$

Let $x \in$ eigen-space(λ).

$$\begin{aligned} Ux = \lambda x \rightarrow x^*U^* &= x^*\overline{\lambda} \\ x^*U^*Ux &= (x\overline{\lambda})\lambda x \\ x^*Ix &= |\lambda|x^*x \\ |x| &= |\lambda||x|, x \neq 0 \\ 1 &= |\lambda| \end{aligned}$$

7. Show that the standard Hermitian inner product on $\mathbb{C}^n$ defines a distance $d(\mathbf{x}, \mathbf{y}) = ||\mathbf{x} - \mathbf{y}||$ on $\mathbb{C}^n$. Give an example of an isometry φ of $\mathbb{C}^n$ such that $\varphi(\mathbf{0}) = \mathbf{0}$ but φ is not $\mathbb{C}$-linear. /3

   Properties of distance
   1) d(x,x)=0, already satisfied
   2) if $x \neq y$, then d(x,y) > 0. Since only 0 has absolute value 0, this property is satisfied. 3) d(x,y)=d(y,x)
   $||x-y||$=$\sqrt{(x-y)\cdot(x-y)} = \sqrt{(y-x)\cdot(y-x)} = ||y-x||$

   4) Triangle Property $d(x,y) \leq d(x,z) + d(z,y)$ Since norms also satisfy the triangle property $|u+v| \leq |u| + |v|$. Then let $|u+v| = |x-y| \wedge |u| = |x-z| \wedge |v| = |z-y|$. So the triangle property is fulfilled.

8. Let φ be an orthogonal transformation of a real inner product space V. Assume that $\varphi^2 = -I$. Show that $\dim V$ is even, say $2n$. Moreover, prove that there exists a dimension n subspace $W \subset V$ and an isometry $\psi : W \to W^\perp$ such that, in the decomposition $V = W \oplus W^\perp$, the operator φ is given by the block matrix /3

$$\begin{bmatrix} \mathbf{0} & -\psi^* \\ \psi & \mathbf{0} \end{bmatrix}$$

   (N.B. The result means that φ can be thought of as multiplication by i on a complex vector space whose real and imaginary parts are W and $W^\perp$.)

   Solution

   If dimension of V is odd, then after applying the transformation twice, the basis can't be left-handed because either the transformation changes signs twice or not at all, and in either case the matrix has to have positive determinant after applying the transformation twice, so the dimension of V can't be odd.

   Now let's prove that the specified subspace exists in the ambient space.

   Claim 1: $\varphi(x) \perp x$.

$$
\begin{aligned}
\varphi^2(x) &= -x, \varphi(-\varphi) = I \\
\varphi^{-1} &= -\varphi = \varphi^T \\
x^T(\varphi^2)^T &= (\varphi^2 x)^T = (-x)^T \\
& x{\cdot}\varphi x \\
& -\varphi^2 x \cdot \varphi x \\
& x^T \varphi^T \cdot -\varphi^2 x \\
& x^T - \varphi \cdot -\varphi^2 x \\
& x^T \varphi^3 x \\
-x^T \varphi x &= x \cdot \varphi x \rightarrow x\varphi x = 0
\end{aligned}
$$

The number can only be equal to both its positive and negative iff the number is 0.

Claim 2: We can construct W.
Pick V_1 and $\varphi(V_1) = V_1^{'}$.
Pick $V_2 \perp V_1, V_1^{'}$, then $\varphi(V_2) \perp V_1$ and $V_1^{'}$, since the inner product is conserved under orthogonal transformations. Keep making pairts until you have n pairs, where 2n is the dimension of V. Now take W to be the subspace defined by taking the span of $V_1, V_2, ..., V_n$. Then φ takes W to $W^{\perp}$, and induces ψ, which is just a restriction of the domain to W and restrction of the codomain to $W^{\perp}$. Since the inner product is conserved in $W^{\perp}$, the distance between points is also conserved and therefore ψ is also an isometry.

Any $x \in V$ can be decomposed into an element in W and $W^{\perp}$. Therefore we just need to figure out where ψ takes the elments from the W. Let's take a basis $v \in W$, and its corresponding pair $v^{'}$.

$$
\begin{aligned}
\psi(v) &= v^{'} \text{ By Defintion.} \\
\psi^*(v^{'}) &= v \text{ Because } \varphi^* = \varphi^{-1}
\end{aligned}
$$

So make the block matrix

$$
\begin{bmatrix} \mathbf{0} & -\psi^* \\ \psi & \mathbf{0} \end{bmatrix}
$$

Which takes $x = w + w^{'}$ to $w' - w$, and $w' - w$ to $-w' - w$, which satisifies the original function.

9. True or false: the sum of two normal operators is normal. Justify. /3

Solution

Take this counterexample. Where A and B, are normal opertors.

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 9 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & -3 \\ 3 & 0 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 5 & -3 \\ 3 & 9 \end{bmatrix}$$

$$(A + B)(A + B)^* = \begin{bmatrix} 34 & -12 \\ -12 & 90 \end{bmatrix}$$
$$(A + B)^*(A + B) = \begin{bmatrix} 34 & 12 \\ 12 & 90 \end{bmatrix}$$
$$(A + B)(A + B)^* \neq (A + B)^*(A + B)$$

So the statement is not true.

10. Show that the space of positive (semi)definite real symmetric matrices is convex. Is the same true with "complex Hermitian" in place of "real symmetric"? /3

Solution

Assume A and B are postive semi definite matrices. Then A and B, satisfy

$$x^T Ax \geq 0 \;\forall x$$

Now we want to show that αA+$(1-\alpha)$B, also satisfies this property.

$$x^T(\alpha A + (1-\alpha)B)x = \alpha x^T Ax + (1-\alpha)x^T Bx \geq 0 \;\forall x \wedge 0 \leq \alpha \leq 1$$

So the real positive matrices are convex. This proof also works for conjugate transposes too.

11. Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$. Find all square roots of A; note that they are all self-adjoint. /3

Solution

Since the trace of the matrix is 6, $\lambda_1 + \lambda_2 = 6$. If we let $\lambda = 1$, $(A - I)$, becomes singular. So the other eigen value becomes 5.

The eigen vector in eigen space of 1, is (1,1), and the other eigen vector is (-1,1).

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sqrt{D} = \begin{bmatrix} \sqrt{5} & 0 \\ 0 & 1 \end{bmatrix}$$

$$\sqrt{A} = P\sqrt{D}P^{-1}$$

12. Find a singular decomposition of the matrix $A = \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix}$. Use it to find $\max_{\|\mathbf{x}\| \le 1} \|A\mathbf{x}\|$ and $\min_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$, as well as the vectors where this maximum and minimum are attained. Describe geometrically the image under A of the closed unit disk in $\mathbb{R}^2$. /3

Solution

$$A^T A = \begin{bmatrix} 2 & 0 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 6 & 13 \end{bmatrix}$$

The eigenvalues of are 16 and 1, therefore the singular values are 4 and 1. Therefore the norm operator is 4, and the minimum length of unit vectors is 1.

The eigen valeus of $A^T A$ are $(1, 2)$ and $(-2, 1)$ for the singular values 4 and 1 respectively. So the vector that achieve the maximum and the minimum are precisely the normalized version of this vectors.

From this we also find V^*.

$$V = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

To W we just do $w_[i] = \frac{1}{\sigma_i} A v_i$.

$$W = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$$

So $A = W\Sigma V^*$, where Σ is a just an operator is the singular values in the diagonal.

13. Prove that the operator norm of a matrix A coincides with the Frobenius norm of A /3 if and only if A has rank at most 1.

Solution

If A has rank 0, there is nothing to prove. The frobenius norm of a matrix is the largest singualr value of the matrix(From the notes). If A has rank one, that means it only has 1, non-zero singular value, therefore that value has to be the one that has to outputed by the frobenius norm.

If the matrix has rank greater than 1, then the frobenius norm and the operator norm can't be equal since the frobenius norm is the sqaure root of the sum of the squares of the singular values, which is greater than the heighest singular value. There the relationship is iff.